

Detection of Land Cover Changes from Khor Al-Dhakhira to Al Khor City Using Spectral Indices Derived from Landsat-8 and Sentinel-2 Data on the Google Earth Engine Platform

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Abstract

The coastal zones of Qatar—particularly the stretch extending from Khor Al-Dhakhira to Al Khor City—exhibit pronounced environmental sensitivity, especially in light of the intensive development projects currently underway. This study seeks to monitor and assess land use and land cover changes within the area over the period 2014–2024. A descriptive-analytical methodology was adopted, utilizing remote sensing techniques through the computation of several spectral indices: the Normalized Difference Vegetation Index (NDVI), the Normalized Difference Water Index (NDWI), the Normalized Difference Built-up Index (NDBI), and the Normalized Difference Soil Index (NDSI). The analysis relied on Landsat-8 and Sentinel-2 satellite imagery, which were processed and interpreted using the Google Earth Engine (GEE) platform. This approach enabled the effective use of pre-built algorithm libraries designed for advanced satellite data processing.

The findings revealed profound land cover transformations, including a considerable reduction in the extent of water bodies and wetlands surrounding Khor Al-Dhakhira, alongside a notable expansion of urban areas and infrastructure. Moreover, the vegetation index analysis indicated a decline in natural vegetation cover in certain locations as a result of urban growth.

Overall, the study underscores the substantial anthropogenic influence on coastal ecosystems and demonstrates the efficiency of employing spectral indices in conjunction with the GEE platform for precise and rapid environmental change detection. These findings are particularly valuable for environmental monitoring and sustainable management of these critical blue carbon habitats under Qatar's National Vision 2030 framework. The results further emphasize the need to integrate such analytical outputs into sustainable urban development planning and to establish long-term monitoring frameworks for natural resources, thereby promoting the resilience and stability of coastal ecosystems.

Keywords : Change Detection, Land Cover, Spectral Indices, Khor Al-Dhakhira, GEE.

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1 Introduction :

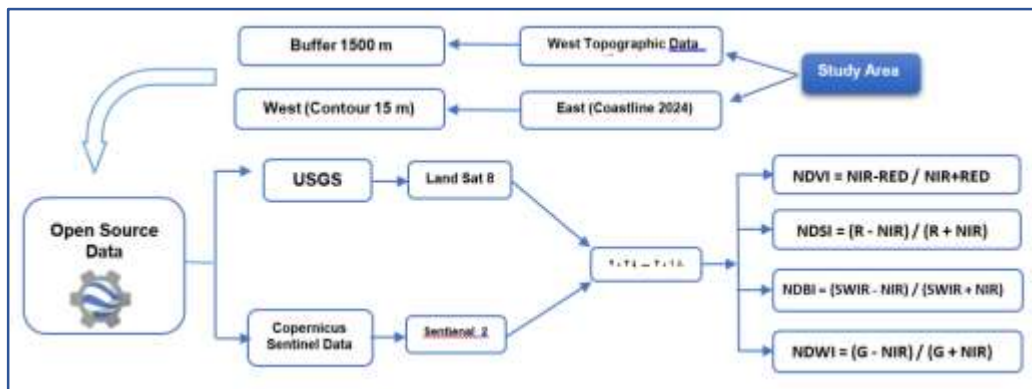
The land cover in Khor Al-Dhakhira, North-east Qatar , has undergone continuous changes from 2018 to 2024 due to natural and human-induced factors such as urbanization, land reclamation, climate change, and agricultural activities. Studying and analyzing these changes over this specific time period is crucial for assessing their impact on the unique ecosystem of this area, which combines marine and terrestrial environments.

Land cover change monitoring serves as a vital tool in environmental studies, enabling quantitative and qualitative assessment of ecological system changes resulting from both human and natural factors. The Google Earth Engine (GEE) platform provides unique capabilities for geospatial data analysis through advanced processing of multi-spectral and multi-temporal satellite imagery such as Landsat and Sentinel-2 data with high precision. This platform employs sophisticated methodologies including spectral analysis using vegetation and water indices, time series analysis, and automated classification techniques that enhance the accuracy of identifying environmental transformations. Practical applications of this scientific approach include tracking urban expansion, evaluating vegetation cover loss rates, and studying water body changes, providing a solid scientific foundation for informed environmental decision-making.

2 Materials and methods

2.1 Work flow

Data selection based on the study area and study period, aimed at analyzing temporal changes, includes Landsat 7 (L7) data from 2001, Landsat 8 (L8) data from 2024 with higher spatial resolution (15 meters), in addition to Sentinel-2 data with a spatial resolution of up to 10 meters for 2015 and 2024 Figure 1 . Using open-source data, primarily the Google Earth Engine (GEE) Platform , the Normalized Difference Vegetation Index (NDVI) was calculated to estimate the increase and decrease in mangrove forest areas in Khor Al Dhakhira, northeast Qatar. The NDVI formula was applied to extract mangrove forests in the study area is $NDVI = \frac{NIR - RED}{NIR + RED}$ (Rouse, J. et.1974).



Using Google Earth Pro and using ArcGIS 10.8 software.

Figure 1 : methodology.

2,2 Data Processing & Methodology

The study leveraged the Google Earth Engine (GEE) cloud computing platform to implement three core spectral indices : the Normalized Difference Vegetation Index (NDVI) for vegetation analysis, the Normalized Difference Water Index (NDWI) for water body identification, and the Normalized Difference Built-up Index (NDBI) for urban feature mapping. These indices collectively enabled comprehensive quantitative assessment of land cover dynamics through GEE's integrated processing algorithms, providing robust spectral discrimination capabilities across the study area.

- The Normalized Difference Vegetation Index (NDVI), developed by Rouse et al. (1974) for vegetation monitoring using near-infrared (NIR) and red (R) spectral bands, is calculated as $(NIR - R) / (NIR + R)$. Healthy vegetation exhibits strong NIR reflectance and red light absorption, yielding positive NDVI values typically ranging from 0.1 to 0.8 (Tucker, 1979), while non-vegetated surfaces show values near or below zero (Jackson & Huete, 1991). This index is fundamental for quantifying vegetation density, photosynthetic activity, and biomass across various ecosystems (Pettorelli & et al., 2005). The practical application of NDVI is demonstrated in a study on the mangroves of Al-Khor, Qatar, where the findings confirm that "the mangroves are in good condition in terms of nitrogen availability and chlorophyll levels, two key factors in the photosynthesis process and are interlinked with each other" (Al-Naimi, Raouf, Al-Ghamdi, & Al-Maadeed, 2016).

-Normalized Difference Water Index (NDWI) :Developed by McFeeters (1996) to monitor water bodies and vegetation moisture, using the green (G) and near-infrared (NIR) bands through the equation : $NDWI = (G - NIR) / (G + NIR)$. The index ranges between -1 and +1, where higher positive values indicate water presence or moisture (Xu, 2006).

-Normalized Difference Built-up Index (NDBI) : Introduced by Zha et al. (2003) to distinguish urban areas from vegetation, using short-wave infrared (SWIR) and near-infrared (NIR) bands : $NDBI = (SWIR - NIR) / (SWIR + NIR)$, Built-up areas show positive values (>0.2), while vegetation exhibits negative values (Estoque & Murayama, 2017).

-Normalized Difference Soil Index (NDSI) : Designed for bare soil detection using red (R) and near-infrared (NIR) bands , $NDSI = (R - NIR) / (R + NIR)$, Bare soils register positive values (>0), while vegetation shows negative values (Bannari et al., 2018).

For vegetation indices such as the Normalized Difference Vegetation Index (NDVI), both sensors utilize the red and near-infrared (NIR) bands. In Landsat 7, the red band is Band 3 (0.63–0.69 μm), and the NIR band is Band 4 (0.77–0.90 μm). In comparison and Landsat 8 uses Band 4 (0.64–0.67 μm) for red and Band 5 (0.85–0.88 μm) for NIR. Although these bands serve similar functions, the narrower and more centered spectral range in Landsat 8 improves sensitivity to vegetation changes and enhances spectral separability between land cover types.

- The spectral bands for calculating the Normalized Difference Vegetation Index (NDVI) were extracted (Van de Griend & Owe, 1993), Where :

Table 1: Spectral Indices in Landsat 8 and Sentinel-2.

Spectral Index	USE	Landsat 8	Sentinel-2	Spatial Resolution
NDVI	vegetation monitoring	NIR – RED / NIR – RED		(L8): 30 m
		B5, B4	B8, B4	(S2):10 m
NDWI	Water body detection & moisture monitoring	Green – NIR / Green + NIR		(L8): 30 m
		B3, B5	B3, B8	(S2):10 m
NDBI	Urban area mapping & built-up land detection	SWIR – NIR / SWIR + NIR		(L8): 30 m
		B6, B5	B11, B8	(S2):20 m
NDSI	dry/exposed soils and wet/organic-covered areas	RED – SWIR2 / RED + SWIR2		(L8): 30 m
		B4, B7	B4, B12	(S2):20 m

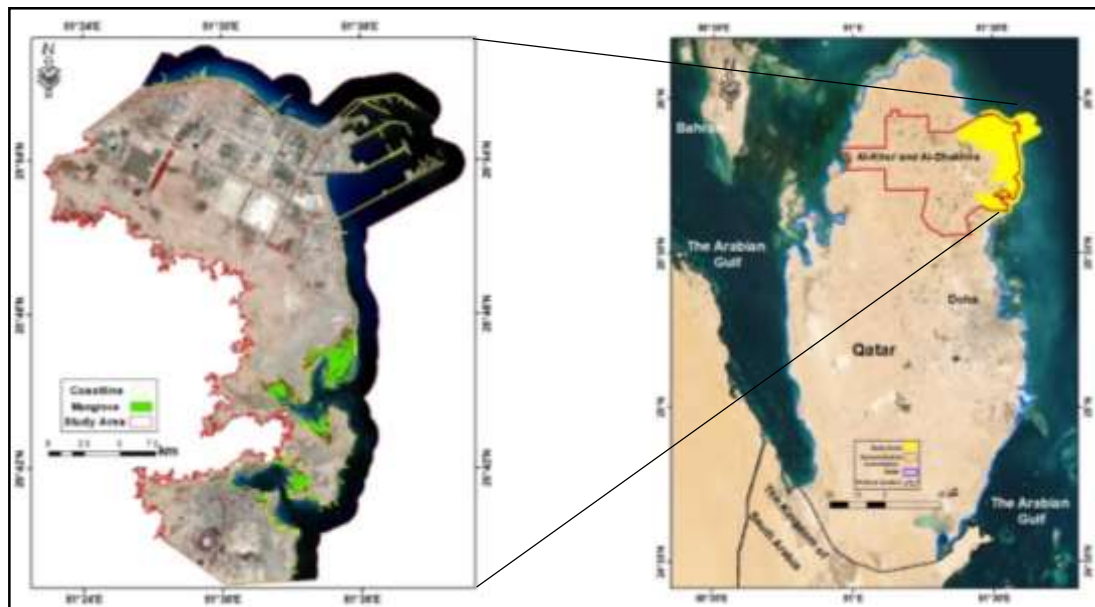
www.usgs.gov/landsat-missions/landsat-8

[chrome-extension://efaidnbmninnbpcjpcglclefindmkaj/https://sentinels.copernicus.eu/documents/247904/685211/Sentinel-2_User_Handbook](https://sentinels.copernicus.eu/documents/247904/685211/Sentinel-2_User_Handbook)

3 Detecting Study area

3,1 in Arc Map

The study area was defined in ArcMap as a shapefile with the following boundaries, It extends westward into the desert hinterland, reaching 15 meters within the administrative boundaries of Al-Khor and Al-Dhakhira. Geographically, it lies between the coordinates 25° 57' 18" North at Khor Al-Dhakhira and 25° 39' 25" South at Al-Khor City. The area is characterized by a distinctive vegetation cover, known as mangrove forests, which stretch along the eastern coast facing the Arabian Gulf. This region is of significant ecological importance due to the presence of mangrove forests that contribute to coastal protection, support biodiversity, and act as carbon sinks, Figure 2.

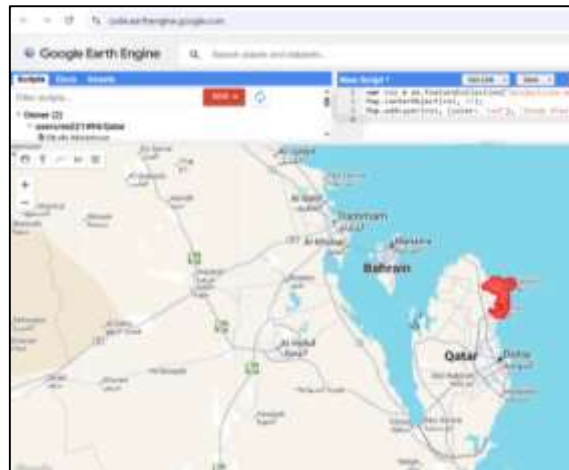


Using Google Earth Pro and using ArcGIS 10.8 software.

Figure 2 : Study area delineation in ArcMap.

3,2 in GEE

The Google Earth Engine platform was employed to estimate mangrove canopy coverage by utilizing satellite imagery from Landsat 7 (ETM+) and Landsat 8 (OLI) spanning the period from 2001 to 2024, (Landsat Collections in Earth Engine,2024) whereas Sentinel-2 imagery was incorporated for the period between 2018 and 2024 (Sentinel Collections in Earth Engine,2024) to take advantage of its higher spatial and temporal resolution, allowing for more detailed and accurate assessment of vegetation dynamics and mangrove health using spectral indices such as NDVI, The programming code used to define the spatial framework of the study area is illustrated in Figure (3).



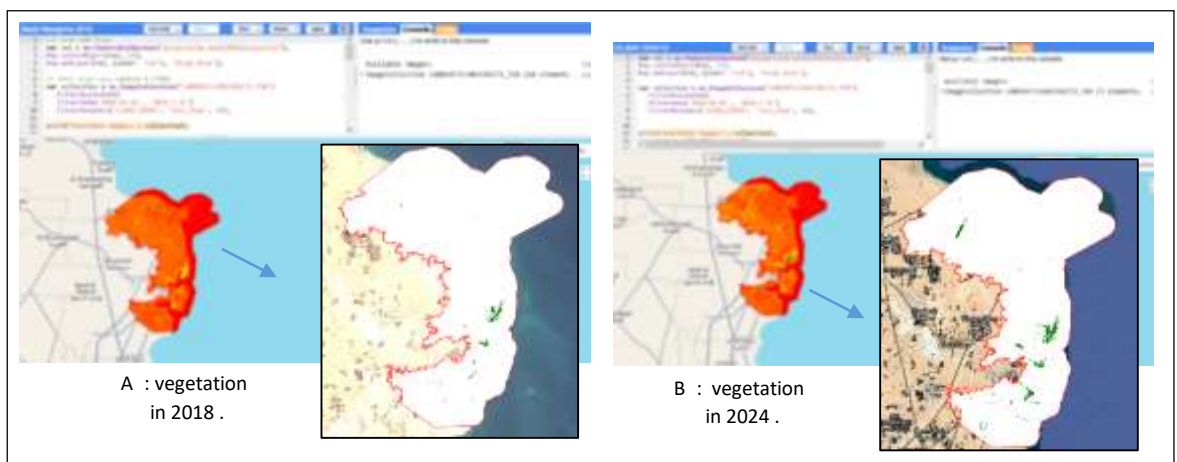
Using Google Earth Engine.

Figure 3 : Script for loading study area in GEE.

4 Discussion

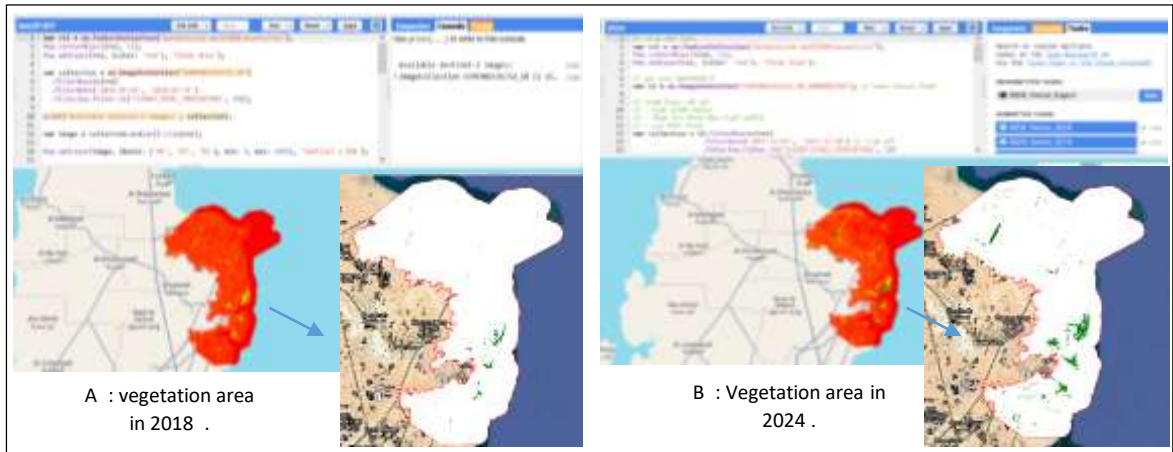
4.1 NDVI (Normalized Difference Vegetation Index).

The Normalized Difference Vegetation Index (NDVI) is one of the most common spectral indices for monitoring vegetation cover, and it can be easily calculated and applied on the Google Earth Engine (GEE) platform. This index relies on the distinctive spectral characteristic of healthy plants, which involves absorbing light in the red region due to chlorophyll and strongly reflecting light in the near-infrared region .



Using Google Earth Engine.

Figure 4 : Estimating NDVI using Landsat 8 .



Using Google Earth Engine.

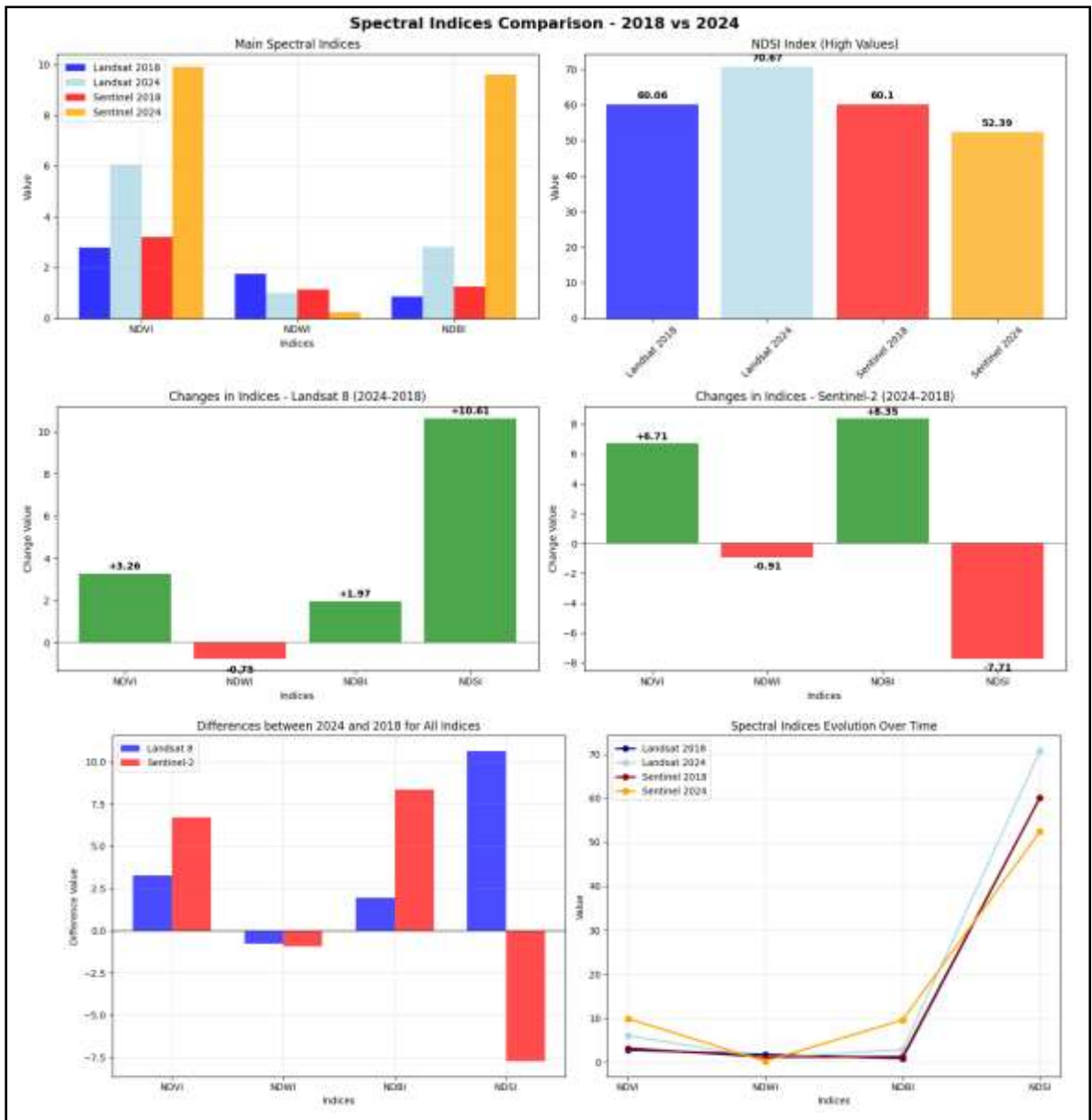
Figure 5 : Estimating NDVI using Sentinel-2 .

Table 2: Assessing Landscape Change Through Spectral Indices Areas

:A Landsat 8 and Sentinel-2 Study from 2018 to 2024.

Spectral Index	Landsat 8		Difference	Sentinel-2		Difference
	2018	2024		2018	2024	
NDVI	NIR – RED / NIR – RED					
	2.78	6.04	+ 3.26	3.19	9.9	+ 6.71
NDWI	Green – NIR / Green + NIR					
	1.74	0.99	- 0.75	1.14	0.23	- 0.91
NDBI	SWIR – NIR / SWIR + NIR					
	0.84	2.81	+ 1.97	1.24	9.59	+ 8.35
NDSI	RED – SWIR2 / RED + SWIR2					
	60.06	70.67	+ 19.89	60.1	52.39	-7.71

Using Google Earth Engine& Arc Map.



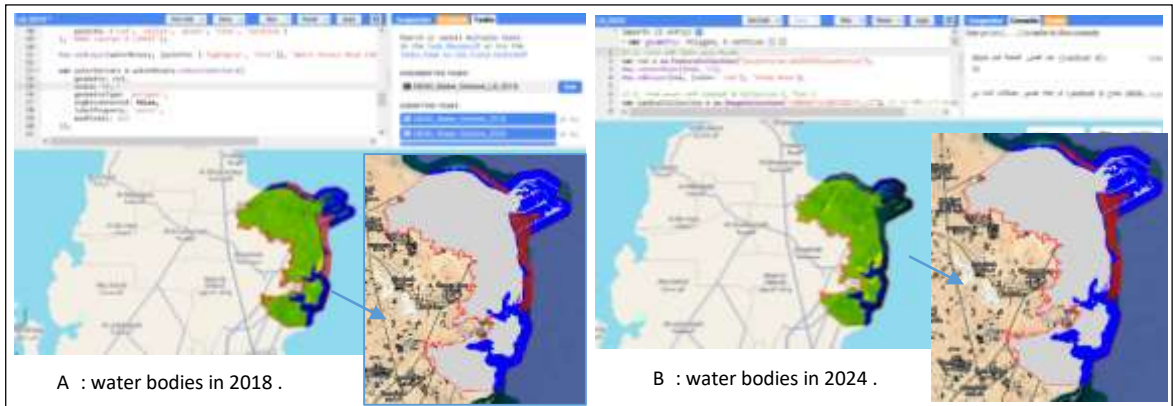
Using Google Earth Engine.

Figur6: Temporal Analysis of Spectral Indices Comparison between Landsat 8 and Sentinel-2 for the Period 2018-2024".

As illustrated in Figure 5, the multi-temporal analysis reveals distinct vegetation growth patterns between the two satellites. Landsat 8 showed an increase from 2.78 to 6.04 in cumulative NDVI values (2018-2024), representing a net gain of 3.26 standardized units after radiometric and atmospheric corrections. In contrast, Sentinel-2 demonstrated significantly higher vegetation detection capacity, recording 6.71 times more growth than Landsat-8. This substantial discrepancy is primarily attributed to Sentinel-2's superior spatial resolution (10m versus 30m), enhanced spectral characteristics, and improved capacity for monitoring fragmented vegetation patterns at finer scales.

4.2 NDWI (Normalized Difference Water Index).

The Normalized Difference Water Index (NDWI) is an effective quantitative tool in the field of remote sensing, primarily used to identify and monitor surface water bodies. The scientific principle of the index is based on the unique optical property of water, which exhibits high absorption of radiation in the Near-Infrared (NIR) band compared to its higher reflectivity in the Green spectral band. This relationship is represented by the mathematical formula: $(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$. The index values range from -1 to +1, where high positive values close to +1 indicate the presence of pure water bodies, while negative values indicate non-aqueous areas such as soil and vegetation. The index features diverse applications including drought monitoring, water resources management, and the study of aquatic ecosystems, making it a valuable tool in hydrological research and environmental studies (McFeeters, 1996).



Using Google Earth Engine.

Figure 7 : Estimating NDWI using Landsat 8 .



Using Google Earth Engine.

Figure 8 : Estimating NDWI using Sentinel-2 .

The values showed a decrease in the NDWI between 2018 and 2024 in both Landsat-8 and Sentinel-2 data, reflecting a decline in water bodies or surface moisture. These results are attributed to the area's potential vulnerability to climatic factors such as rising temperatures or drought, or to human activities leading to the depletion of water resources, such as unregulated agricultural expansion or increased withdrawal from surface and groundwater sources.

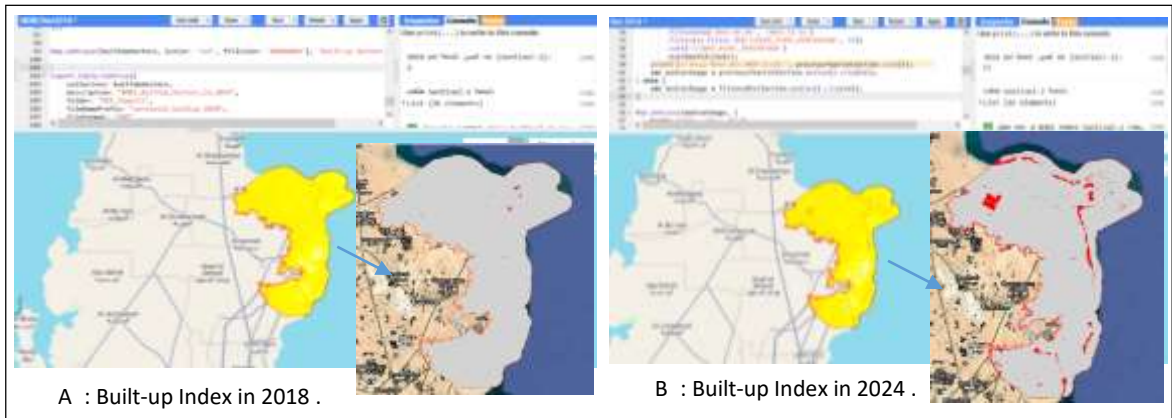
4.3 NDBI (Normalized Difference Built-up Index) .

Is a remote sensing index that automatically maps urban areas using satellite imagery. It calculates the normalized difference between Short-Wave Infrared (SWIR) and Near-Infrared (NIR) reflectance: $(SWIR - NIR)/(SWIR + NIR)$.Built-up areas show positive values (0.2 to 1) due to high SWIR reflectance, while vegetation and water yield negative values, The Built-up Index is based on the unique spectral response of built-up areas and their surroundings. Built-up areas are characterized by higher reflectance in the shortwave infrared (SWIR) band compared to the near-infrared (NIR) band, which is the opposite of vegetation response. This property is exploited by the NDBI to automatically map urban areas, (Zha,Y., Gao, J., & Ni, S.2003).



Using Google Earth Engine.

Figure 9 : The script for Calculating the NDBI using Landsat 8 .



Using Google Earth Engine.

Figure 10 : The script for Calculating the NDBI using Sentinel-2 .

The results revealed a clear increase in NDBI values between 2018 and 2024 in both Landsat-8 and Sentinel-2 imagery, which clearly indicates an expansion of the urban footprint and urban development during this period. This rise reflects increased construction activities and infrastructure expansion, often consistent with population growth and accelerated urbanization, which may consequently lead to the decline of some natural ecosystems due to urban pressure.

4.4 NDSI (The Normalized Difference Soil Index).

Is a specialized remote sensing index used for soil characterization. It leverages the distinct spectral reflectance properties of materials in the red (RED) and short-wave infrared (SWIR2) regions. The index is mathematically defined as $(RED - SWIR2) / (RED + SWIR2)$. This calculation effectively distinguishes dry, exposed mineral soils, which reflect highly in the SWIR2 band yielding lower NDSI values, from wet soils or those covered by organic matter, which absorb SWIR2 radiation and consequently produce higher NDSI values. Therefore, NDSI serves as a quantitative tool for monitoring

soil moisture content, exposure, and organic cover in various environmental applications .

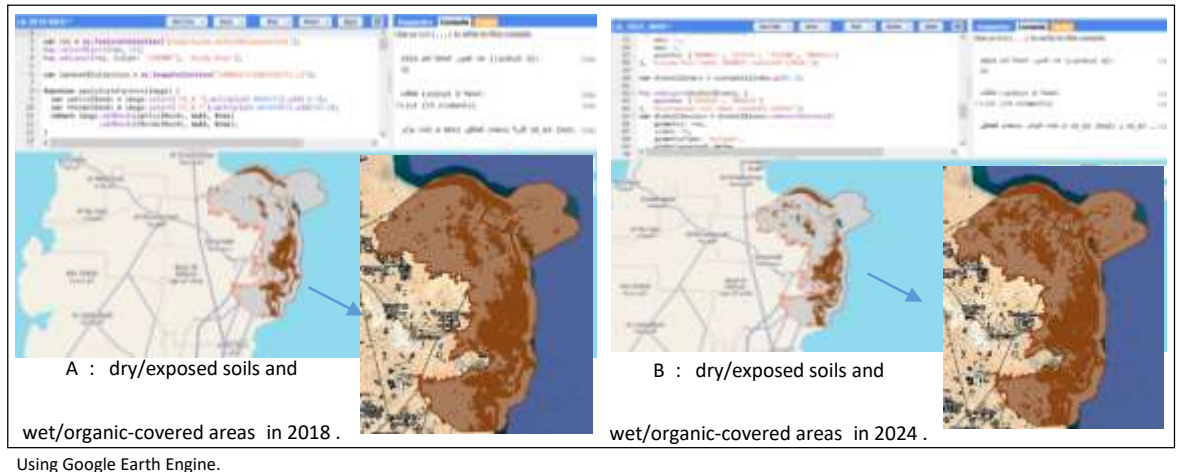


Figure 11 : The script for Calculating the NDSI using LandSat-8 .

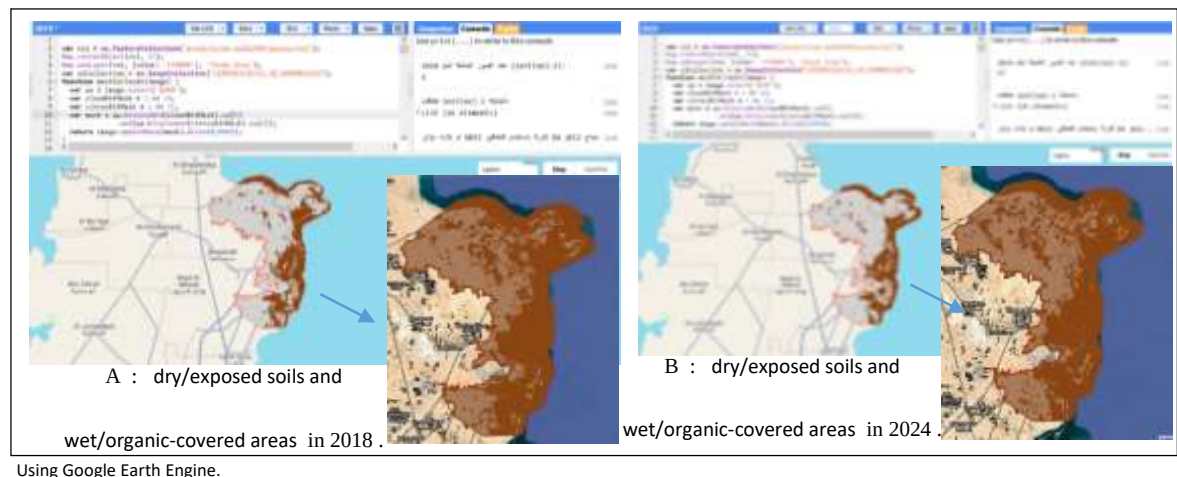


Figure 12 : The script for Calculating the NDSI using Sentinel-2 .

The study results revealed a divergence in trends between Landsat-8 data, which recorded an increase in NDSI values, and Sentinel-2 data, which exhibited a decrease. This discrepancy can be attributed to differences in the spatial and temporal resolution of the two satellites, in addition to varying atmospheric conditions during imaging. However, the overall trend of rising NDSI values in some scenes may indicate an increase in exposed or reflective surfaces, such as dry soil and sand. This aligns with the recorded decrease in NDWI and increase in NDBI, collectively reflecting an environmental shift toward relative aridity and growing human activities in the study area.

Through an integrated comparison of the four spectral indices derived from both Landsat-8 and Sentinel-2 data, it is evident that the study area underwent clear environmental and spatial transformations between 2018 and 2024. The Normalized Difference Vegetation Index (NDVI) showed a notable upward trend, indicating an increase in biomass and an improvement in overall vegetation cover. This may be attributed to the expansion of agricultural activities or the implementation of environmental policies aimed at greening. In contrast, the Normalized Difference Water Index (NDWI) exhibited a clear downward trend, indicating a decline in surface water resources or a decrease in soil moisture content, potentially due to climatic impacts or increased human water consumption.

Furthermore, the Normalized Difference Built-up Index (NDBI) recorded a significant increase during the same period, clearly

pointing to an expansion of the urban footprint and increased urban sprawl at the expense of agricultural or natural lands. This is an indicator of growing urban pressure that could contribute to future environmental changes if the expansion is not accompanied by sustainable planning measures.

. The Normalized Difference Soil Index (NDSI) showed a divergent trend between Landsat and Sentinel data; however, the general trend indicates a relative increase in exposed and dry surfaces, reinforcing the hypothesis of receding moisture and rising surface aridity in the area.

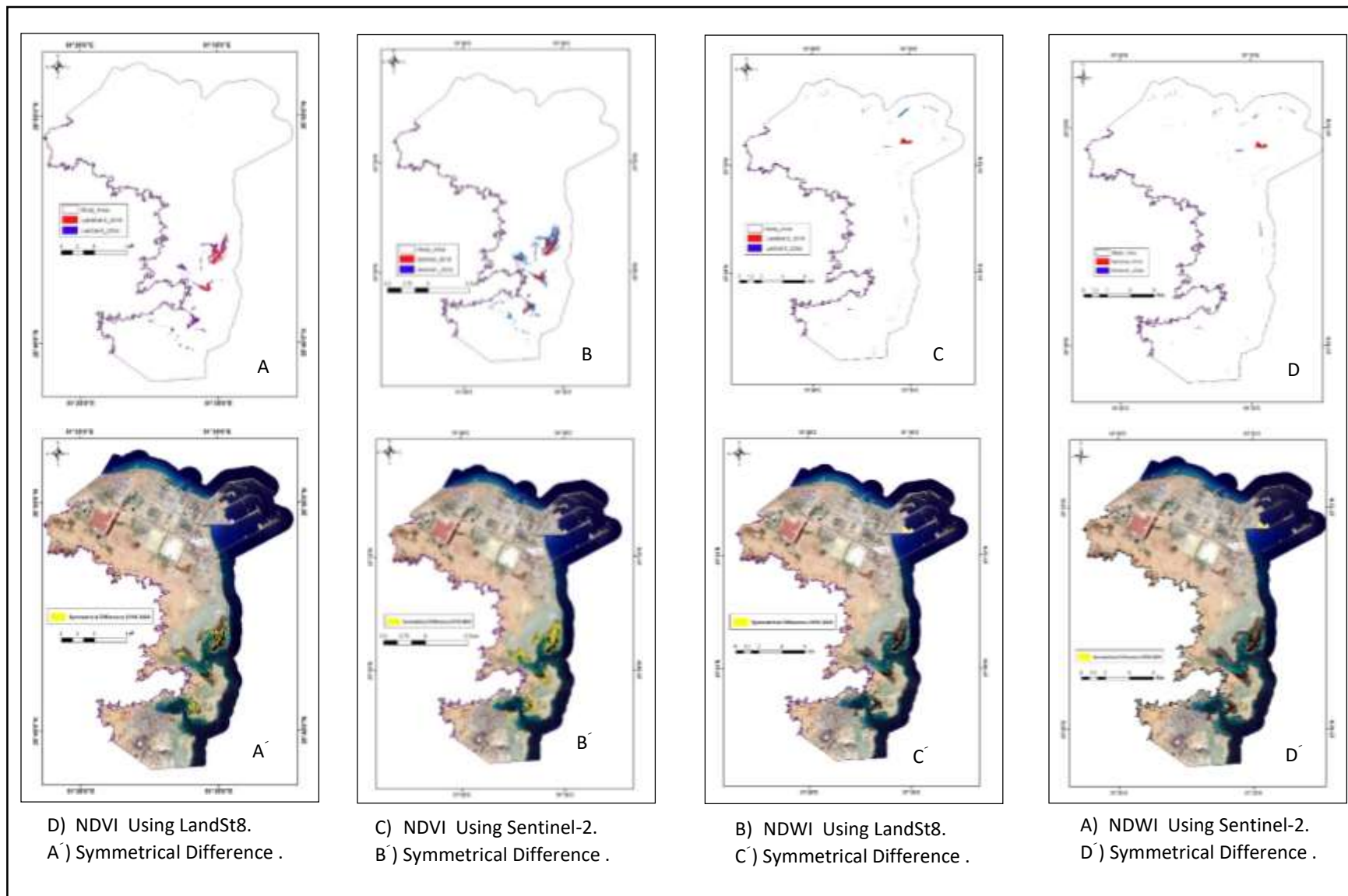
Based on the foregoing, it can be concluded that the area experienced a complex environmental shift during the study period, characterized by increased vegetation and urban cover alongside declining water resources and increased surface aridity. This environmental pattern reflects the dynamic interaction between human activities and climatic factors, necessitating the enhancement of integrated natural resource management policies to ensure the sustainability of the ecological balance in the coming years.

5 Calculate Change Detection:

Change Detection is a fundamental process in remote sensing that involves identifying and analyzing differences in the state of an object or phenomenon by observing it at different times (Habibullah Abbasi, 2021) . This technique is crucial for monitoring the Earth's dynamic surface, providing objective data that supports decision-making in urban planning, environmental protection, disaster management, and climate research . The core challenge of

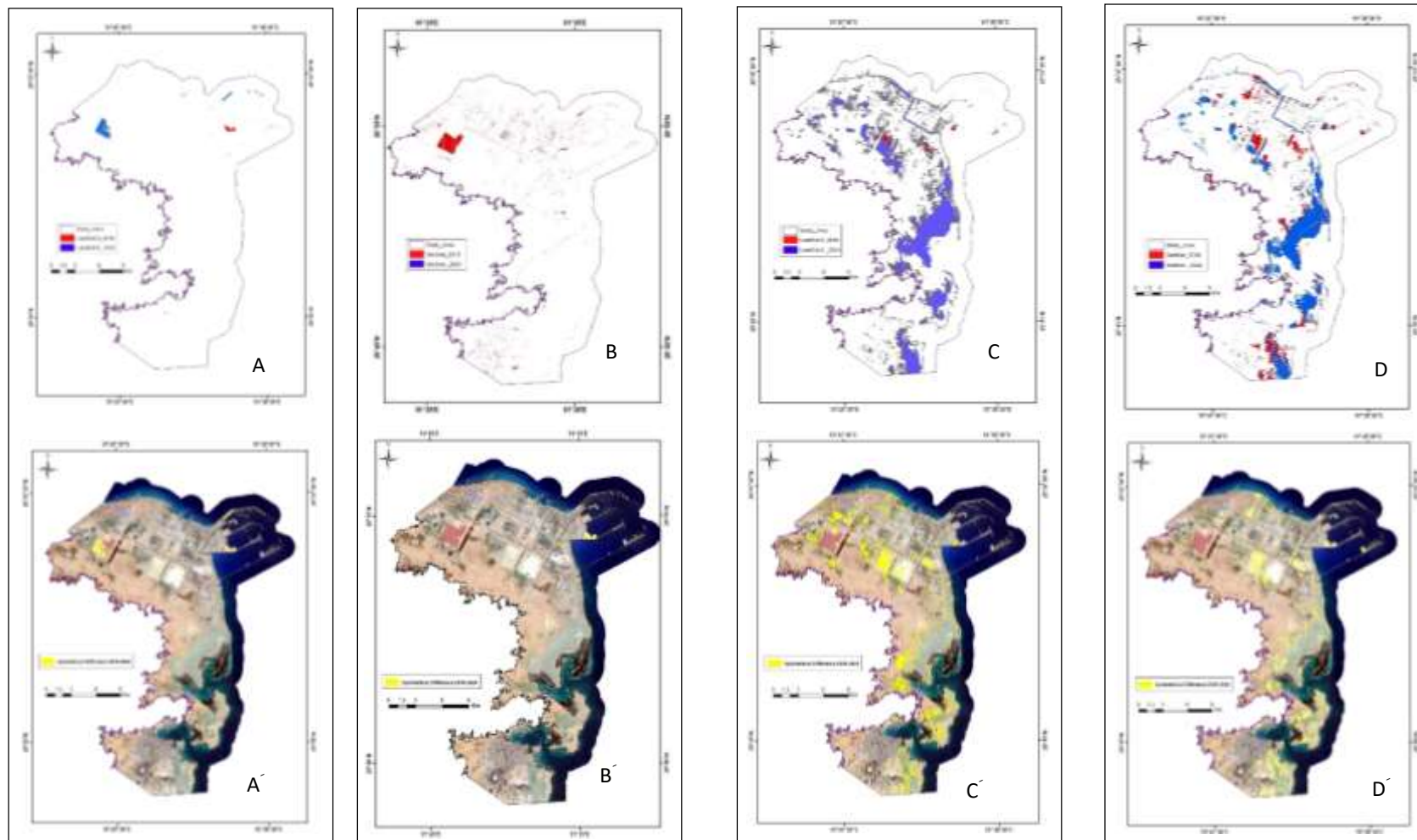
CD lies in distinguishing genuine changes on the ground from spurious differences caused by variations in sensor characteristics, atmospheric conditions, illumination (Guangliang Cheng , 2024).

Change detection studies utilizing remote sensing techniques represent a fundamental methodological step for monitoring and analyzing spatiotemporal transformations of the Earth's surface. This study is grounded in conducting a precise comparison between the performance of the Landsat 8 and Sentinel-2 satellites during the period from 2018 to 2024, a critical timeframe marked by accelerated environmental and climatic changes.



Using ArcGIS 10.8.

Figure 13 : Change Detection Result for Satellite Spectral Indices using LandSat8 and Sentinel-2 from 2018 - 2024 .



A) **NDBI** Using **LandSat 8**.

A') Symmetrical Difference .

B) **NDBI** Using **Sentinel-2**.

B') Symmetrical Difference .

C) **NDSI** Using **LandSat 8**.

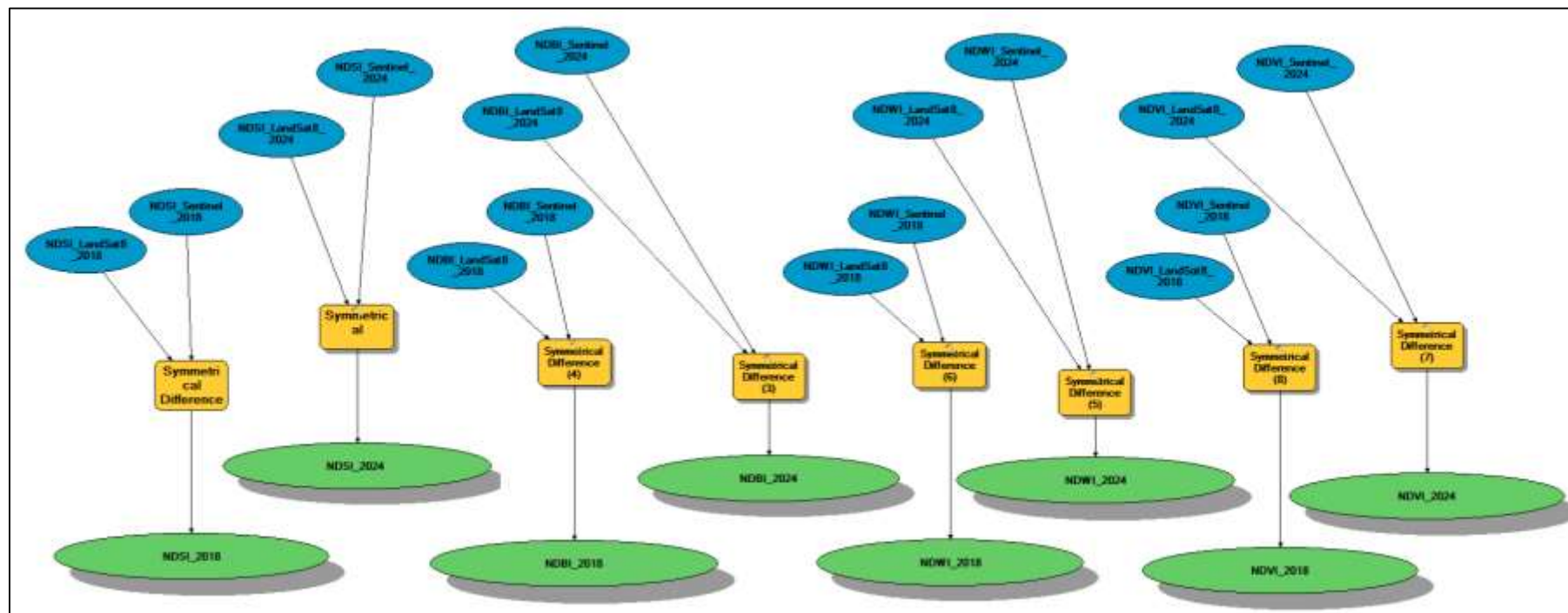
C') Symmetrical Difference .

D) **NDSI** Using **Sentinel-2**.

D') Symmetrical Difference .

Using ArcGIS 10.8.

Figure 14 : Change Detection Result for Satellite Spectral Indices using LandSat8 and Sentinel-2 from 2018 - 2024 .



Using ArcGIS 10.8.

Figure 15 : Change Detection for Satellite Spectral Indices from 2018 – 2024.

Table 3 : Spatial Change Assessment of Spectral Indices Using Multi-Satellite Data (2018-2024).

Spectral Index	2018		Difference in Resolution (L8-Sen2)	2024		Difference in Resolution (L8-Sen2)
	LandSat 8	Sentinel 2		LandSat 8	Sentinel 2	
NDVI	2.78	3.19	+ 0.41	6.04	9.9	+ 3.86
NDWI	1.74	1.14	- 0.6	0.99	0.23	- 0.76
NDBI	0.84	1.24	+ 0.4	2.81	9.59	+ 6.78
NDSI	60.06	60.1	- 0.04	70.67	52.39	- 18.28

Using ArcGIS 10.8.

The analytical methodology relied on processing spatial data within the ArcMap software environment to derive a set of key biophysical spectral indices: NDVI (Normalized Difference Vegetation Index), NDWI (Normalized Difference Water Index), NDBI (Normalized Difference Built-up Index), and NDSI (Normalized Difference Snow Index). The comparison aims to evaluate the degree of consistency and divergence in the accuracy of each satellite in detecting these changes, taking into account differences in technical specifications such as spatial resolution (30 meters for Landsat versus 10 meters for Sentinel in specific bands) and temporal recurrence (16 days versus 5 days). This comparative analysis has contributed to identifying the strengths and limitations of each space mission, thereby providing a comprehensive perspective on the suitability of their data for sustainable monitoring systems and advanced geospatial applications.

The presented figure illustrates a systematic analytical workflow for comparing spectral indices (NDSI, NDBI, NDWI, NDVI) derived from two distinct satellite data sources (Landsat-8 and Sentinel-2) across two temporal periods (2018 and 2024). The application of a Symmetrical Difference operation to each individual index yields a set of binary change maps. These maps explicitly identify the specific geographical locations that underwent change, whether detected over the temporal domain (between the two dates) or attributable to the sensor/platform characteristics (between the two data sources).

conclusion :

Figure 13 and Table 6 collectively illustrate the spatial and temporal variations of satellite-derived spectral indices (NDVI, NDWI, NDBI, and

NDSI) from 2018 to 2024, using multi-sensor data from Landsat-8 and Sentinel-2 processed in ArcGIS 10.8. The figure demonstrates the analytical workflow applied for change detection, highlighting the use of the symmetrical difference method to quantify the degree and direction of change between two temporal datasets. Each spectral index from both sensors was analyzed separately, and their differences were integrated to generate a final change detection output that identifies significant environmental alterations.

The values in Table 6 reveal notable variations between 2018 and 2024. NDVI values increased significantly for both Landsat-8 (from 2.78 to 6.04) and Sentinel-2 (from 3.19 to 9.9), suggesting an overall improvement in vegetation density or regrowth in certain patches. NDWI values decreased considerably—from 1.74 to 0.99 (Landsat-8) and from 1.14 to 0.23 (Sentinel-2)—indicating reduced surface water coverage and declining moisture conditions, potentially due to land reclamation or coastal modification. NDBI values rose sharply, especially for Sentinel-2 (from 1.24 to 9.59), reflecting intense urban and infrastructural expansion, particularly in coastal and industrial zones. Meanwhile, NDSI displayed mixed trends—an increase for Landsat-8 (60.06 → 70.67) but a decrease for Sentinel-2 (60.1 → 52.39)—implying variability in surface exposure and sediment redistribution. The discrepancies between Landsat-8 and Sentinel-2 readings are mainly attributed to differences in spatial resolution (30 m vs. 10–20 m), acquisition timing, and radiometric corrections. Sentinel-2's finer resolution allows more precise detection of micro-scale spatial changes, while Landsat-8 captures broader regional patterns. Methodologically, the symmetrical difference approach ensures

that only genuine change pixels are retained, minimizing classification noise. This integration of multi-sensor data strengthens the temporal consistency and reliability of the change detection results. These variations in values are primarily attributed to differences in spatial resolution between the two satellites, which will be elaborated in detail in the subsequent section :

- In terms of the spatial resolution of the bands used to calculate the spectral indices (NDVI, NDWI, NDBI, NDSI), technical data analysis demonstrates a clear superiority of the Sentinel-2 satellite for most of these indices.
- For the vegetation cover (NDVI) and water content (NDWI) indices, Sentinel-2 uses the Red and Near-Infrared (NIR) bands at a spatial resolution of 10 meters, while Landsat 8 provides these bands at only 30 meters. This means that Sentinel-2 offers a significantly higher capability to discern fine details in vegetation and small water bodies.
- Regarding the NDBI (Built-up areas) index, which relies on the Short-Wave Infrared (SWIR) band, Sentinel-2 provides this band at 20 meters compared to 30 meters in Landsat 8, allowing for a more precise analysis of urban expansion.
- Regarding the NDSI dry/exposed soils index, which combines the visible Green band and a Short-Wave Infrared (SWIR) band, the difference is even more pronounced. Sentinel-2 uses the Green band at 10 meters and the SWIR band at 20 meters, whereas Landsat 8 provides both bands at 30 meters.

Finally, the combined use of Landsat-8 and Sentinel-2 datasets provides a robust and complementary framework for long-term environmental change monitoring in coastal and urbanizing regions.

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كشف التغيرات في الغطاء الأرضي من خور الذخيرة إلى مدينة الخور باستخدام المؤشرات الطيفية من بيانات لاندسات-8 و سينتال-2 باستخدام منصة Google Earth Engine .

ملخص البحث :

تتميز المناطق الساحلية في قطر، لا سيما نطاق الدراسة الممتد من خور الذخيرة إلى مدينة الخور، بحساسية بيئية عالية في ظل ما تواجهه المنطقة من مشاريع تنموية، لذا تسعى الدراسة إلى رصد وتحليل التغير في استعمالات الأراضي والغطاء الأرضي بالمنطقة خلال الفترة 2014-2024م، معتمدةً على التحليل الوصفي باستخدام تقنيات الاستشعار عن بُعد من خلال حساب المؤشرات الطيفية : الدليل الطبيعي للغطاء النبات (NDVI)، الدليل الطبيعي للمياه (NDWI)، الدليل الطبيعي للمناطق العمرانية (NDBI)، والدليل الطبيعي للترية (NDSI)، اعتماداً على بيانات LandSat8 و Sentinel_2 ثم معالجة البيانات وتحليلها باستخدام منصة Google Earth Engine (GEE)، الأمر الذي مكّن من الإستفادة من مكتبات الخوارزميات الجاهزة المخصصة للمعالجة المتقدمة للبيانات الفضائية.

كشفت النتائج عن تحولات جذرية في الغطاء الأرضي، تمثلت في تراجع ملحوظ في مساحة المسطحات المائية والأراضي الرطبة حول خور الذخيرة، مقابل توسع واضح في الرقعة العمرانية والبُنى التحتية، كما كشف تحليل الدليل النباتي عن تناقص في مساحة الغطاء النباتي الطبيعي ببعض المناطق، نتيجة للتمدد العمراني.

وتؤكد الدراسة من خلال هذه النتائج على حجم التأثير البشري في النظم الإيكولوجية الساحلية، كما تبرهن على كفاءة استخدام المؤشرات الطيفية ومنصة GEE في رصد التغيرات البيئية بدقة وسرعة، ولهذه النتائج أهميتها في سياق الرصد البيئي والإدارة المستدامة لنظم الكربون الأزرق الحرجة، مع التوصية بضرورة الإفادة من مخرجات الدراسة في صياغة خطط تنموية مستدامة واعتماد آليات للمراقبة المستدامة للموارد الطبيعية، وذلك تماشياً مع رؤية قطر الوطنية 2030، مما يسهم في الحفاظ على استقرار هذه النظم البيئية.

الكلمات المفتاحية : كشف التغيرات، الغطاء الأرضي، المؤشرات الطيفية، خور الذخيرة، GEE .